

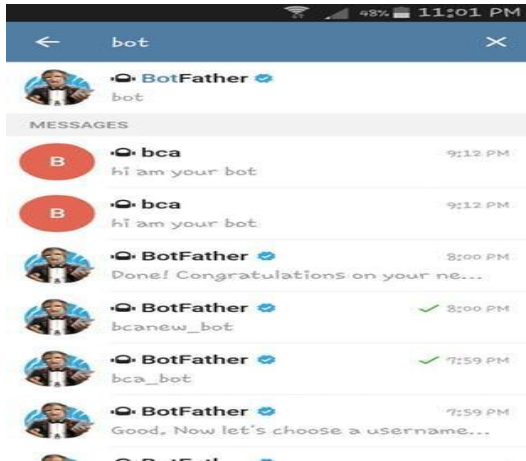
PRACTICAL - 3

Controlling Raspberry Pi with Telegram.

Step 1: Open Telegram app in your system or mobile

Open Telegram app in your system or mobile

Start "BotFather"



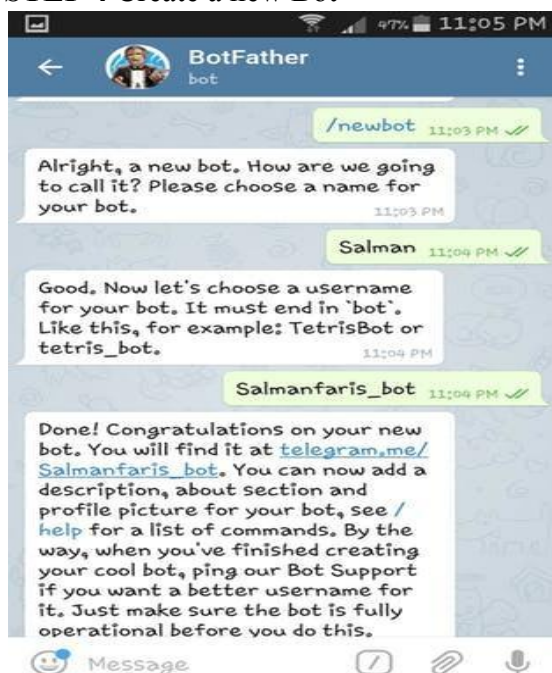
Step 2: Open "BotFather"



STEP 3: /start



STEP 4 Create a new Bot



STEP 5:- Obtain access token



STEP 6:- Install "Python Package Index"

sudo apt-get install python-pip

```
Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Wed Jan  4 16:25:26 2017 from 192.168.100.4
pi@raspberrypi:~ $ sudo apt-get install python-pip
Last login: Wed Jan  4 16:25:26 2017 from 192.168.100.4
pi@raspberrypi:~ $ sudo apt-get install python-pip
Reading package lists... Done
Building dependency tree
Reading state information... Done
python-pip is already the newest version.
0 upgraded, 0 newly installed, 0 to remove and 81 not upgraded.
pi@raspberrypi:~ $
```

STEP 7:- Install "telepot"

(To download telepot create a virtual environment)

1) Installing the venv module:-

sudo apt install python3-venv

2) Creating a new virtual environment. Myenv is user-defined name

python3 -m venv myenv

3) Activate the Virtual Environment

source myenv/bin/activate

4) It should look like :- (myenv) pi@raspberrypi:- \$

5) Install telepot in virtual machine:-

```
pip install telepot
```

6) pip install RPi.GPIO

STEP 8:-

1) Clone the git

```
git clone https://github.com/salmanfarisvp/TelegramBot.git
```

2) Go into the cloned repository

```
cd TelegramBot
```

3) Edit the code to add your bot token

```
nano telegrambot.py
```

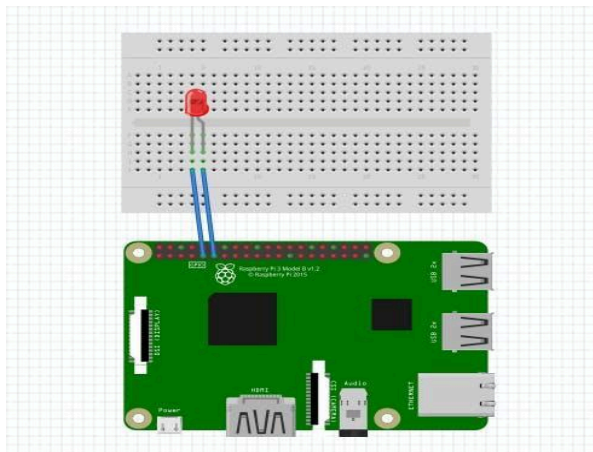
4) After this the .py file will open. Paste the bot token here

```
bot = telepot.Bot('your_bot_token')
```

5) Run the Code

```
python telegrambot.py
```

STEP 9:- All set, now time to connect the Pi and LED.
Connect LED to Pi



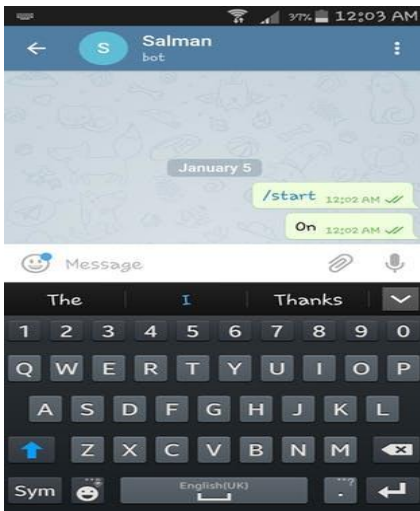
Step 10: Send Command
Start our Bot



START



Send "on" & "off"





Look at your Pi, you can see the LED on and off when you send "on" and "off" to our bot.

Code:

(This code will be displayed after the command:- nano telegrambot.py)

```
import sys
import time
import random
import datetime
import telepot
import RPi.GPIO as GPIO

#LED
def on(pin):
    GPIO.output(pin,GPIO.HIGH)
    return
def off(pin):
    GPIO.output(pin,GPIO.LOW)
    return
# to use Raspberry Pi board pin numbers
GPIO.setmode(GPIO.BOARD)
# set up GPIO output channel
GPIO.setup(11, GPIO.OUT)

def handle(msg):
    chat_id = msg['chat']['id']
    command = msg['text']

    print 'Got command: %s' % command

    if command == 'on':
        bot.sendMessage(chat_id, on(11))
    elif command == 'off':
        bot.sendMessage(chat_id, off(11))

bot = telepot.Bot('Bot Token')
bot.message_loop(handle)
print 'I am listening...'
```

```
while 1:  
    time.sleep(10)
```

STEP 11:- To deactivate virtual environment

```
deactivate
```